

1 • Shared evidence: the cold case

The cold case

Name _____ Date _____

Keep this page beside your chosen response route. All numerical examples below are constructed sample data, not observations from your class.

D1 • Same room

Smooth metal and wood blocks both measured 20 °C. A warmer hand felt the metal as colder.

C1 • First claim

“The metal must have started colder because it felt colder.”

C3 • Delivery claim

“A foam jacket makes cold and is always the best jacket.”

D2 • Melting clue

Equal 20 g ice pieces, both starting at about 0 °C, sat on room-temperature trays. After 5 minutes: metal set-up 9 g melted; plastic set-up 3 g melted. Tray dimensions matched; masses differed.

C2 • Second claim

“The metal set-up transferred energy to the ice faster in this test.”

Delivery test: constructed sample data. Identical sealed bottles, equal fill, same 20 °C room, all starting at 8 °C. Jackets fitted fully; jacket thickness was not matched.

Jacket	Start / °C	After 20 min / °C
None	8	14
Cotton sleeve	8	12
Foam sleeve	8	10

W1. Which claims can you keep, reject or improve? Explain one using D1 or D2.

W2. Which jacket limited warming most? Name one check needed before claiming a material is always best.

2 • Supported route: choose and explain

Use page 1. Point, circle, draw or dictate; the science goal is the same.

S1. The bottle starts at 8 °C. The room is 20 °C. Which is warmer?

A. Bottle

B. Room

S2. Draw an energy arrow between ROOM and BOTTLE. Label the warmer end.

S3. Which final temperature shows the coldest bottle after 20 minutes?

A. No jacket: 14 °C

B. Cotton: 12 °C

C. Foam: 10 °C

S4. Complete: I choose _____. It reached _____ °C, compared with _____ °C without a jacket. It _____ energy transfer into the bottle. Word bank: slows / stops forever.

S5. The two blocks were both 20 °C. Why did metal feel colder? Use: hand / energy / faster.

S6. Name or circle one useful next check: repeat the test / guess from jacket colour. Explain your choice.

3 • Standard route: delivery recommendation

Use page 1. The table repeats constructed sample data from page 1. Complete one evidence-backed decision for the delivery team.

N1. Draw and label the net energy-transfer direction between a 20 °C room and an 8 °C bottle.

Jacket	Start / °C	20 min / °C	Rise / °C
None	8	14	
Cotton	8	12	
Foam	8	10	

N2. Complete the rises. Show one subtraction, then compare foam with no jacket.

N3. Recommend a jacket. Explain its effect on energy transfer using two numerical values.

N4. Explain why the 20 °C metal may feel colder than the 20 °C wood to a warmer hand.

N5. Give one limitation of the jacket evidence and one precise improvement.

4 • Stretch route: challenge the claim

Use page 1. Your task is to justify a useful decision without claiming more than the data show.

T1. Calculate all three temperature rises. Recommend a jacket and explain the direction and rate of energy transfer.

T2. The foam sleeve was 10 mm thick; the cotton sleeve was 2 mm thick. Constructed test details. Can this establish that foam material is better at every thickness? Explain and redesign this comparison.

T3. Now put 40 °C water inside an insulated bottle in the same 20 °C room. Draw energy's direction and explain how the same jacket changes cooling. This is a thought experiment only.

T4. Evaluate: "The ice result proves conductivity alone caused the difference." Use one limitation from D2 and propose one improvement.

5 • Exit and scientific reflection

Answer independently. A drawing or spoken response recorded by an adult is welcome.

E1. Room 20 °C; bottle 8 °C. Draw and label the net energy-transfer arrow.

E2. What does a thermal insulator do?

E3. Give numerical evidence for the foam-jacket recommendation from page 1.

E4. What extra measurement or comparison would you ask the class to investigate next?

Reflection. One idea I changed because of evidence was ...
